# Electric Potential Is The Energy Of a Unit Positive Charge In An E-Field

E-Field PhET Simulation


SYNOPSIS

We explored the electric field in the last PhET simulation lab. You should recognize that **a field is a cue for force**. Specifically, the field lines tell you how an object — a mass in a gravitational field, a charge in an electric field — will accelerate, because it knows the force acting on it. This note is about the other way of looking at the same situation.


## Two world's views

Remember in mechanics that physicists have two "world's views"? One is the **force** perspective. The other is the **energy** perspective. We can analyze the behavior of an object using either, or both. However, I told you that they prefer the energy perspective if they have the choice, because the math is simpler: **force is a vector**, requiring three equations for three dimensions, whereas **energy is a scalar**, so it is just one equation for an object.

So now, let's talk about the energy of a charge in a field! But first, here is a reminder about work (energy), $W$ (J): $W = Fd$, where $F$ is the force (N) and $d$ (m) is the distance over which the force is applied. So work done in a gravitational field is $W_G = F_g d = mgh$, where $F_g = mg$, $m$ is the mass (kg), $g$ is the acceleration due to gravity (9.8 m/s$^2$), and $h$ is the distance over which the gravity force held sway.

## A new term — potential

At this point, I need to introduce a new term, **potential**. **This is not a new concept!** Just like the field is a force-like quantity (when you see "field," think "force"), the potential is an energy-like quantity, so **when you see potential, you should think "energy."**

The field appears in the force equation: $F_g = mg$, where $g$ is the gravitational field, and $F_E = qE$, where $E$ is the electrostatic field. The potential appears in the energy equation. So for gravity,

$$W_G = F_g d = mgh = mU_G$$

where $W_G$ is the gravitational potential energy (as opposed to just potential), and $U_G$ is the gravitational potential, which is just $gh$, **the field multiplied by distance!** For electrostatic force,

$$W_E = F_E d = qEd = qV_E$$

where $W_E$ is the electrostatic potential energy and $V_E$ is the **electric potential**, which is just $Ed$, the electric field multiplied by distance. We will just drop the subscript E in $V_E$ and call this **potential**.

You are already very familiar with $V$: it has its own unit, the **volt** (V), and, yes, it's the same as "voltage" volts as you see in batteries! So you have been using the electrostatic potential to gauge your battery level — voltage goes down as batteries wear out, or as their energy content decreases over time. It's not the actual energy measured in J, but a very good indication of how much energy you have left in the battery. Refer to your Reference Table for "V" on p. 4; it is denoted as **"potential difference."** This is completely equivalent to electrostatic potential or voltage.

### How physicists define it

Finally, I want to describe how physicists define electric potential:

$$V = W/q$$

the same equation as $W_E = qV_E$, where I just drop the subscript E and rearrange for $V$. In other words, the electrostatic potential, or voltage, is **the energy per charge** — the energy in each coulomb of charge (J/C, joules per coulomb).

> TRY THE ARITHMETIC ON IT
>
> So if you have **2 V** of potential and **0.5 C**, you have **1 J**. If there is **4 J** of energy in the charge in a field, and there is **2 C**, then you have **2 V** of potential.

We could do this with gravitation, too, but we just didn't — perhaps there were no compelling reasons for us. Why for electrostatics? **Here, it's really useful for electricity!** In electricity, we have voltages and other parameters to describe electric circuits, and electrostatic potential or potential difference (completely equivalent to voltage, $V$) is a crucial quantity, rather than energies in joules or forces in newtons. More on that soon.

For completeness, let me add the definition of the electric field: $E = F_E/q$, which reads as **force per charge**, or force experienced by each coulomb of charge.

PERFORMANCE EXPECTATIONS

1. **Describe** electrostatic potential as energy in 1 coulomb of charge.
2. **Describe** the potential of a charge next to a single charge.
3. **Describe** the potential of a charge between lines of opposite charges by recognizing the fields between the lines.
4. **Describe** equipotential lines.

### Instructions and questions

In this second part of the "Charges and Fields" PhET simulation, you will explore the potential of a charge (energy in a coulomb of charge) in an electric field. First, you have just a single charge generating a spherical field. Second, you will make lines of charges, one with positive and one with negative charges, and observe the fields and potentials between and outside the lines of charges.

1. Jump back into the simulation. Check the **Grid** and **Voltage** boxes.

## Part I — potentials in single positive and negative charges

2. Use the **voltmeter** to determine the electric potential (an "energy"-like quantity represented by just a number) at each location, labeling each spot as you go. **Include the sign of the potential.** Repeat the procedure with a negative source charge.
3. What is the sign of the potential?
4. How does the potential value change with the distance from the charge?

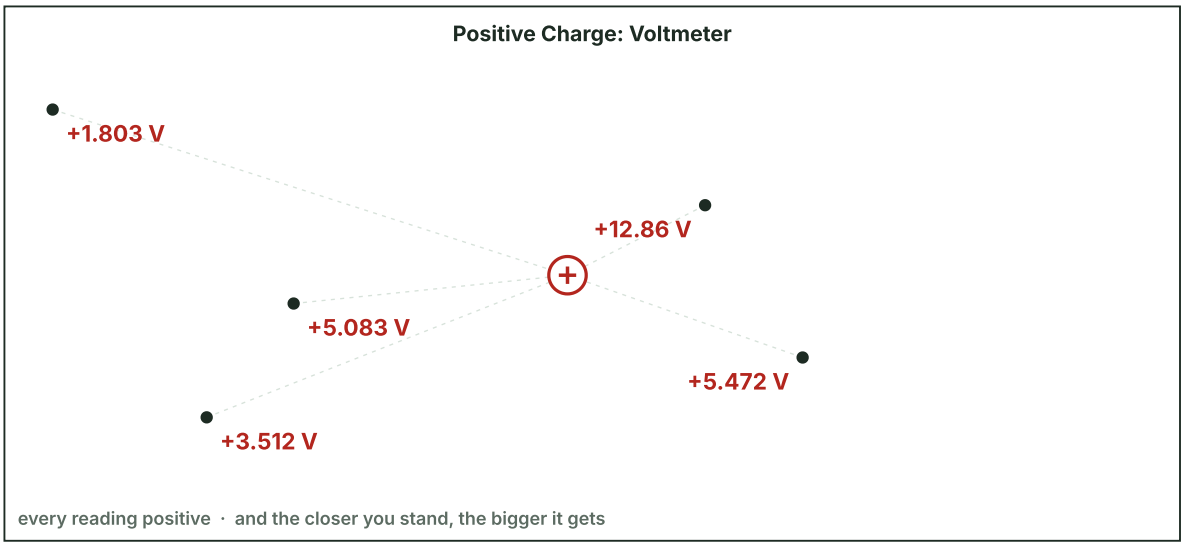



**Figure 1.** The five readings taken in class, at the five places they were taken. **Every one is positive**, and the one nearest the charge is far and away the biggest — +12.86 V against +1.803 V out at the corner. Potential does not point anywhere; it is just a number at each spot.

5. What are the **similarities and differences** in the values of the potentials between the positive and negative single charges?

## Part II — potential in a dipole (one positive and one negative charge in close proximity)

6. To create a dipole, place one positive and one negative charge on the grid, vertically aligned, with a reasonable distance between them.
7. Use the voltmeter to measure the electric potential along locations approximately **equal distances from the charges**. What can you say, generally, about the value of the electric potential equal distance from the opposite charges?
8. **Draw the line of zero potential** on your field map below.

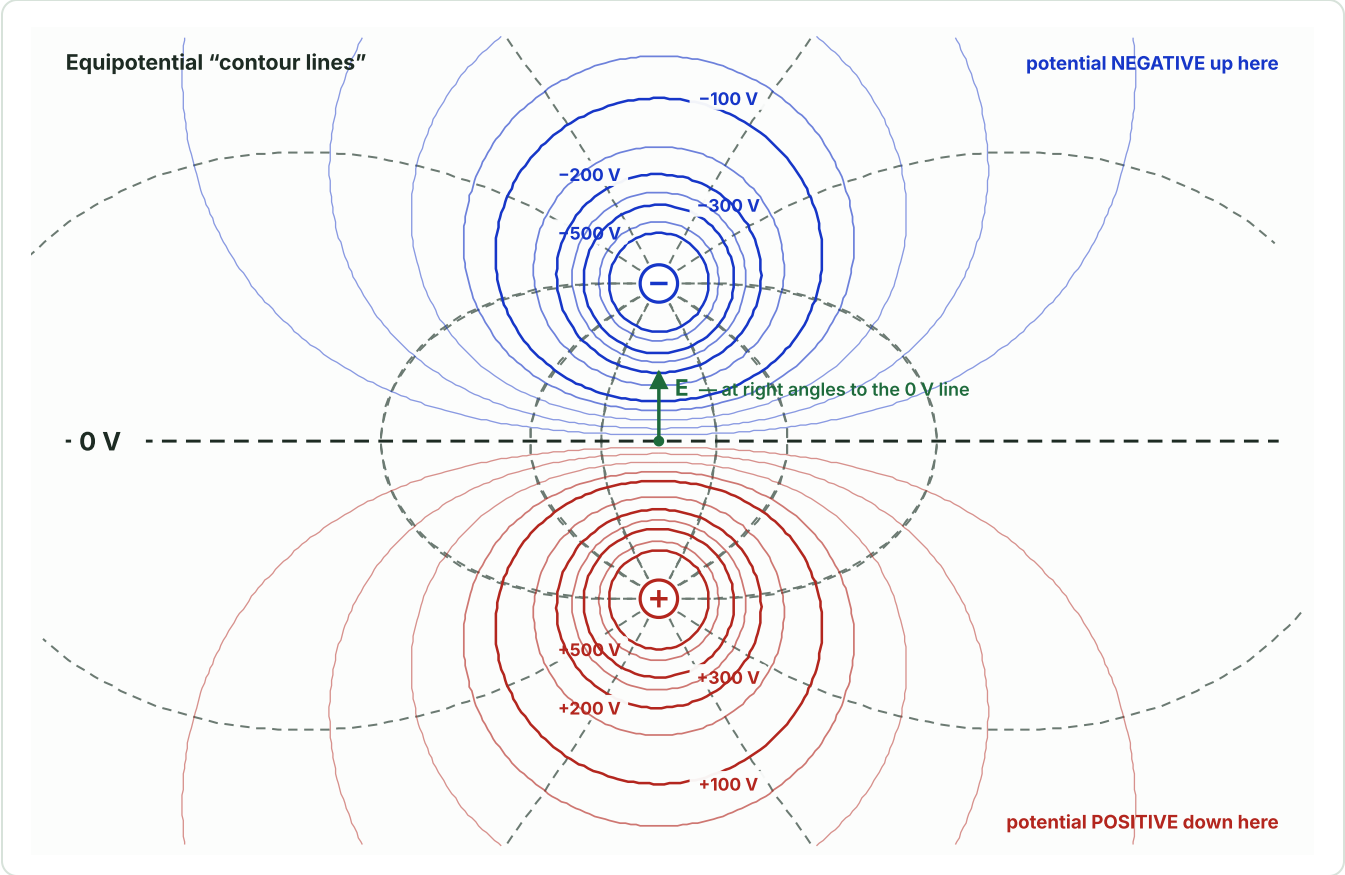



**Figure 2. Equipotential “contour lines.”** Exactly like the contour lines on a map — every point on one of them is at the same potential, and the value is written on it. They crowd together close to a charge, where the potential changes fast, and spread out as you go away. **The 0 V line is dead straight**, exactly midway. And look at the green arrow: **the field crosses the equipotential at right angles**, pointing from the + charge toward the − one — downhill, always. The faint dashed curves are the field lines, and they cut every contour square.

9. Grab the sensor and move it around to find a location or locations **where the field is zero**. Can you find it?

10. How do you explain the existence of **zero electric potential** where the **electric field is not zero**?

11. How does the value of the potential change with the distance from the charges?

12. With the charges close enough to touch each other, place the voltmeter nearer the edge of the field (making it easier to see the pattern). What happens to the voltmeter reading as you are at the edge of the field, and what does this reading imply about how the potentials from positive and negative charges add?

## Part III — potential in two like charges (two negative or two positive charges in close proximity)

13. Place two like charges on the grid vertically aligned with a reasonable distance between them. Use a voltmeter and find the locations of **zero potential**. Can you find it? Explain the result.

14. Draw the electric field lines. Grab a sensor and move it around. **Where is the location of the zero field?** Indicate it on the diagram. Explain the result.

### The same three cases, as graphs

Here is the potential plotted along the line that runs straight through the two charges. $V_1$ is the potential from charge 1 alone, $V_2$ from charge 2 alone, and what the voltmeter reads is the two of them added — **added as numbers**, which is the whole point.

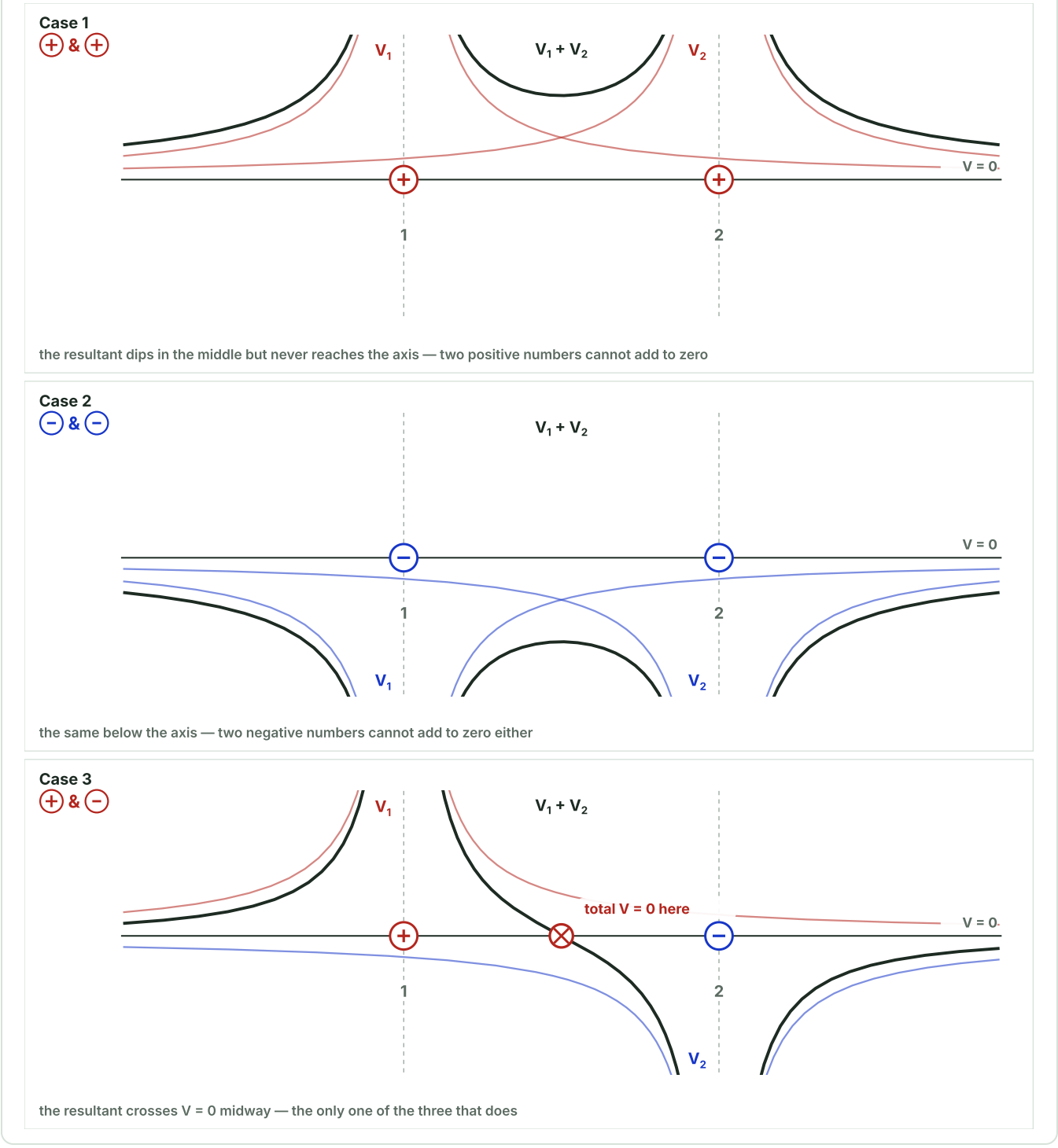



**Figure 3. Up means positive potential; the horizontal line is V = 0.** Each curve runs away to infinity at its own charge and decays toward zero far away. **Case 1: both curves sit entirely above the line. Case 2: both entirely below.** Two numbers of the same sign cannot add to zero, so on neither of those plots can the total ever reach the line — which is exactly the answer to the question above. **Only Case 3, the dipole, crosses zero**, and it crosses midway between the charges.

**The thick black line is the one that matters:** $V_1 + V_2$, which is what the voltmeter actually reads. The thin curves are what each charge would do on its own. In Cases 1 and 2 the thick line dips toward the axis between the charges and then **pulls away again without ever reaching it** — you can watch two same-sign numbers failing to cancel. In Case 3 it goes straight through, at the ⊗.

**15 — OPEN QUESTION, WE DO THIS ONE TOGETHER**

Move one charge and place it **atop the other**. What happens to the field and the potential? Explain the results.

Careful here. These are **like** charges, not a dipole — so before you answer, decide what two vectors pointing the *same* way do when you add them, and what two numbers of the *same* sign do.

## Part IV — potential in two line charges of opposite signs

16. Make two lines of charges, one consisting of positive and one negative. Grab a sensor and map the electric field **between** the line charges and **outside** the line charges. How does the field between the line charges differ from that outside them?

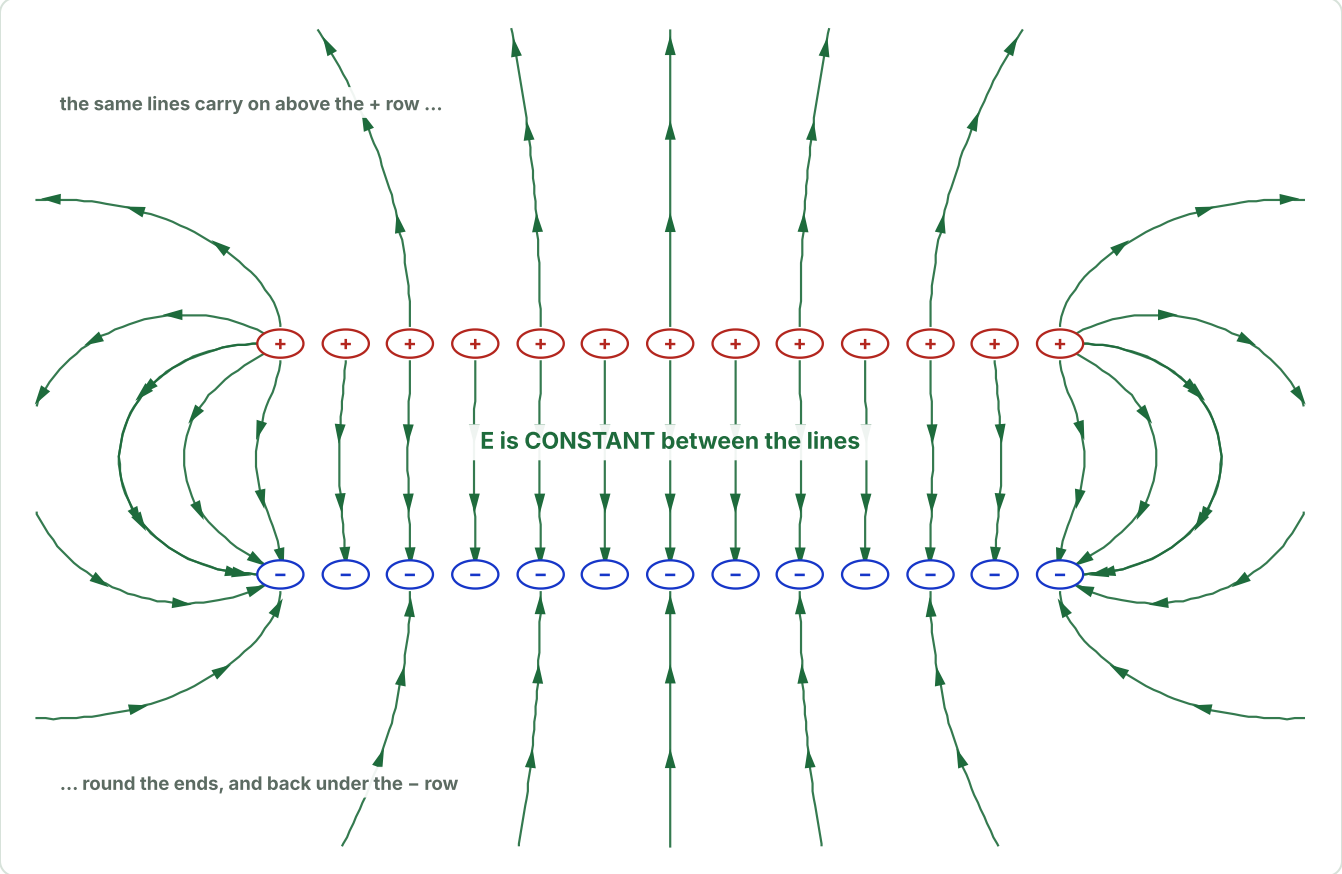


**Figure 4.** Two line charges. **Between them the lines are straight, parallel and evenly spaced**, running + to − — that is what a constant field looks like, and it is the one picture to carry into Module 4, where this geometry comes back as the capacitor.

**Now follow a line past the rows.** The field above the + row and below the − row is not something separate: it is the **same lines carrying on**, up off the top, out round the ends, and back underneath. Notice they point **up in both places** — away from the + row above it, toward the − row below it — and that they are sparse out there compared with the crowd between the rows, which is the field being weak.

17. Use the voltmeter and map the potential between the line charges and outside the line charges. How does the potential between the line charges differ from outside the line charges?

**THE PUNCHLINE**

One is a vector. The other is just a number.

Every answer in this note comes from that. A dipole has **no** zero-field point but a whole **line** of zero potential. Two like charges have **no** zero potential anywhere but a zero-field **point** dead in the middle. You do not have to remember the four cases — you have to remember which quantity is a vector.

## Summary

**YOUR FIVE — NOT MINE**

List **five things** that you learned about electric potential.  Left blank on purpose — write your own.

1. —
2. —
3. —
4. —
5. —

One of mine, if you want a model: **equipotential lines are like the contour lines on a map, representing regions of equal potential.**

## Looking ahead

You now have both halves: a **force** picture and an **energy** picture of the same charges, done with algebra.  **Note 5** does this same subject again with **calculus** — reading the field straight off a contour map, and getting the potential of a spread-out charge with one scalar integral instead of three vector ones.  **Note 8**, later in the module, puts field and potential side by side against the gravitational versions you already know, and asks what Newton would have made of all this.